25DCE026

Department of Computer Engineering

Practical - 11

Practical Title: Student Management CRUD with Search and Filter

Problem Definition: Create a student management module to add, view, edit, delete, search and filter student records. All database operations must use MySQLi prepared statements and display clear success/failure messages.

Theory :

1. Are SELECT, INSERT, UPDATE, and DELETE operations working?

Yes. All CRUD operations use parameterized MySQLi statements. Records can be dynamically created (with uniqueness checks), retrieved (with dynamic filters), modified (pre-populated), and soft-deleted or restored.

2. Is search by name/email/course implemented?

Yes. Fuzzy search matches across full names, emails, and enrollment numbers using SQL `LIKE ?` bindings, with dropdown filters for course and active/deleted status.

3. Are delete operations confirmed before execution?

Yes. Deletion triggers a JavaScript `confirm()` dialog to prevent accidental record removal.

4. How are Intermediate and Advanced extensions implemented?

Intermediate: `export_csv.php` streams student data to the client using PHP `fputcsv()`. **Advanced:** Soft deletion updates `status` to `deleted`, preserving referential integrity with a restore option.

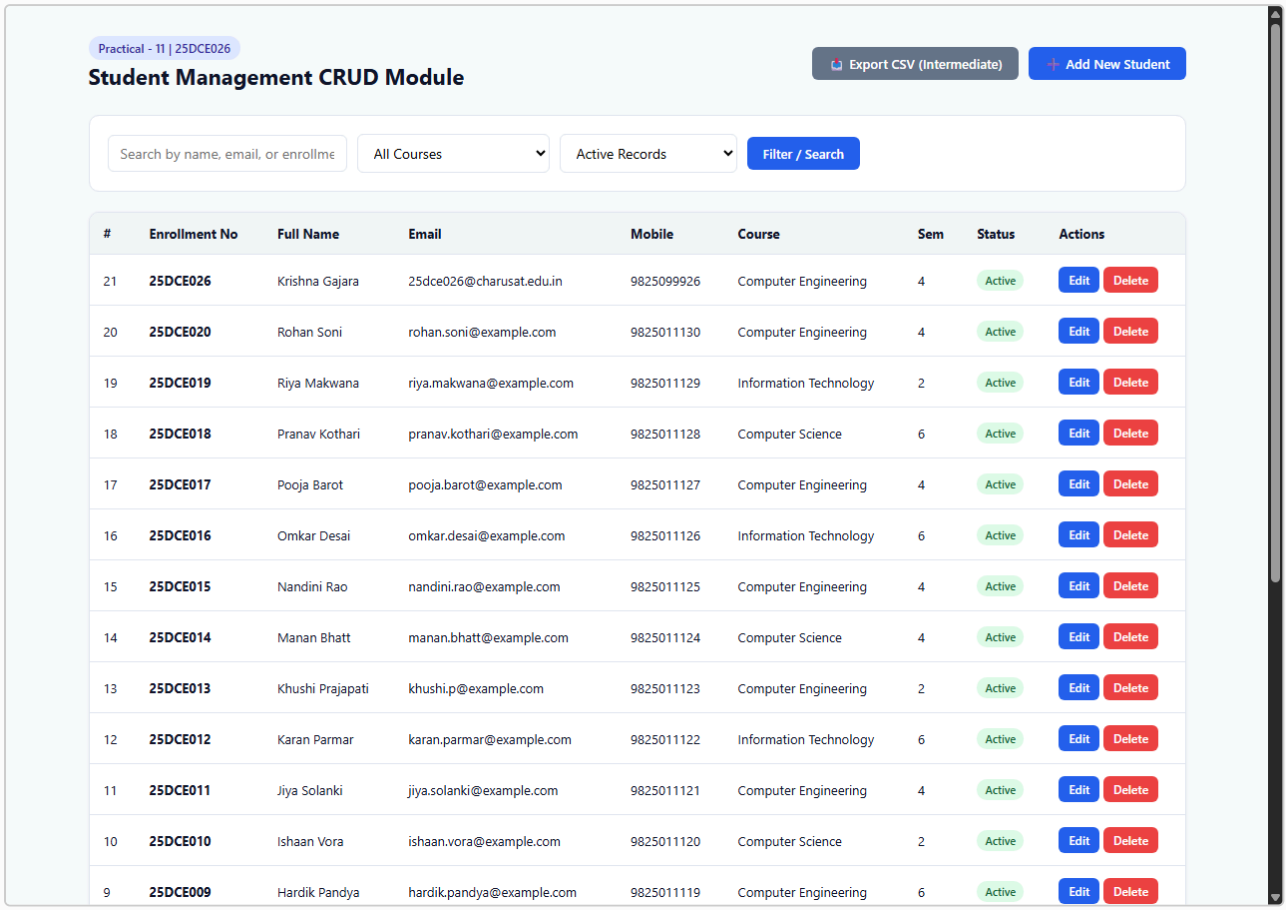
Implementation / Output / Result

```
// Parameterized Multi-Field Search Query
$query = "SELECT * FROM students WHERE status = ? AND (full_name LIKE ? OR email LIKE ? OR
enrollment_no LIKE ?)";
$stmt  = $conn->prepare($query);
$stmt->bind_param("ssss", $status, $search_term, $search_term, $search_term);
$stmt->execute();
$result = $stmt->get_result();
```

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Output 1: Student Management Table with 21 Seeded Records, Multi-Field Search & Filter

Live browser screenshot: Full data table with Enrollment No, Name, Email, Course, Semester, Status columns and Edit/Delete actions.



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Student Management CRUD Module

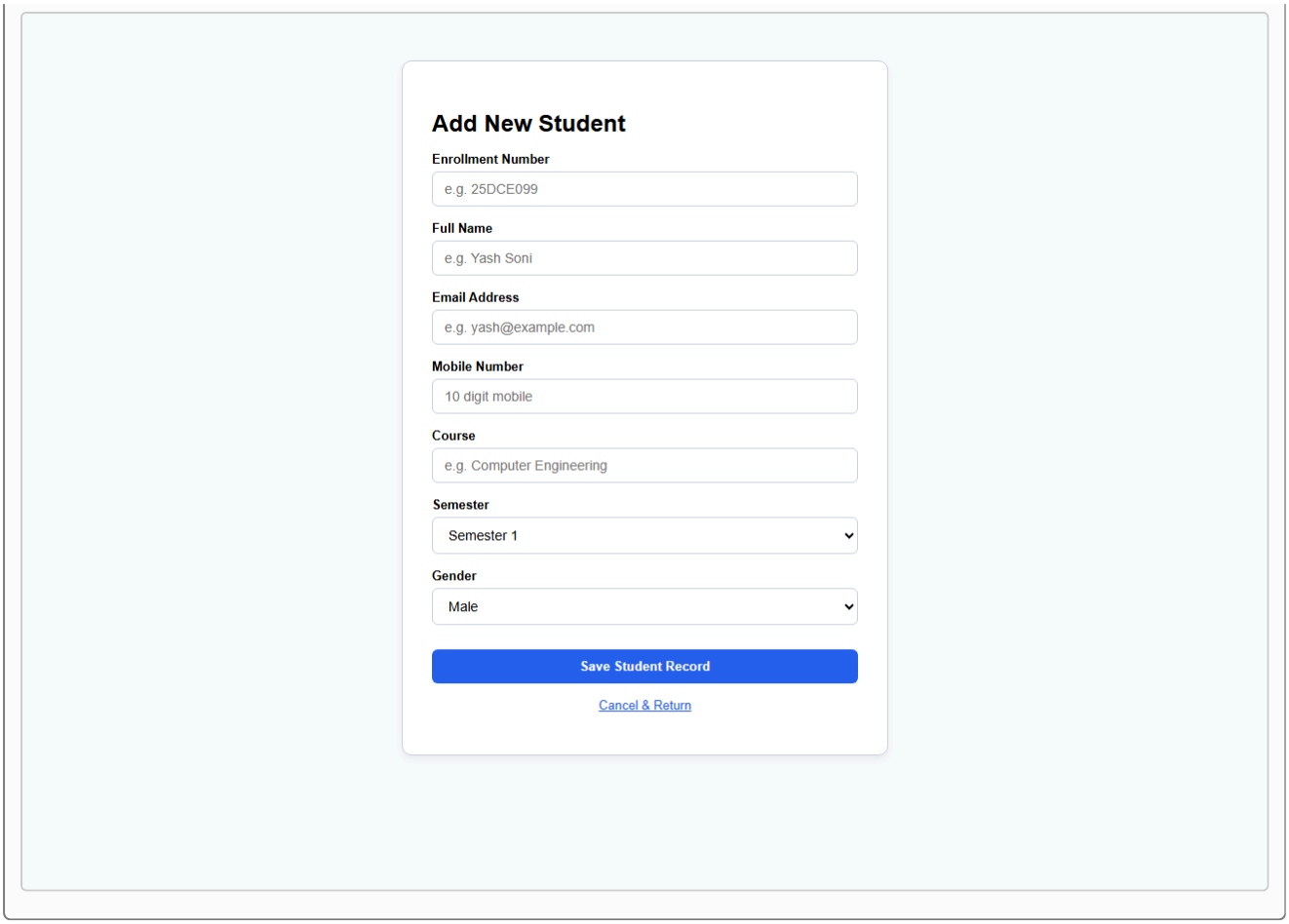
Export CSV (Intermediate) | Add New Student

Search by name, email, or enrollme | All Courses | Active Records | Filter / Search

#	Enrollment No	Full Name	Email	Mobile	Course	Sem	Status	Actions
21	25DCE026	Krishna Gajara	25dce026@charusat.edu.in	9825099926	Computer Engineering	4	Active	Edit Delete
20	25DCE020	Rohan Soni	rohan.soni@example.com	9825011130	Computer Engineering	4	Active	Edit Delete
19	25DCE019	Riya Makwana	riya.makwana@example.com	9825011129	Information Technology	2	Active	Edit Delete
18	25DCE018	Pranav Kothari	pranav.kothari@example.com	9825011128	Computer Science	6	Active	Edit Delete
17	25DCE017	Pooja Barot	pooja.barot@example.com	9825011127	Computer Engineering	4	Active	Edit Delete
16	25DCE016	Omkar Desai	omkar.desai@example.com	9825011126	Information Technology	6	Active	Edit Delete
15	25DCE015	Nandini Rao	nandini.rao@example.com	9825011125	Computer Engineering	4	Active	Edit Delete
14	25DCE014	Manan Bhatt	manan.bhatt@example.com	9825011124	Computer Science	4	Active	Edit Delete
13	25DCE013	Khushi Prajapati	khushi.p@example.com	9825011123	Computer Engineering	2	Active	Edit Delete
12	25DCE012	Karan Parmar	karan.parmar@example.com	9825011122	Information Technology	6	Active	Edit Delete
11	25DCE011	Jiya Solanki	jiya.solanki@example.com	9825011121	Computer Engineering	4	Active	Edit Delete
10	25DCE010	Ishaan Vora	ishaan.vora@example.com	9825011120	Computer Science	2	Active	Edit Delete
9	25DCE009	Hardik Pandya	hardik.pandya@example.com	9825011119	Computer Engineering	6	Active	Edit Delete

Output 2: Add New Student Form with Validation

Live browser screenshot: Add Student form with all required fields — enrollment number, full name, email, course, semester, gender.



GitHub Repository Link:
https://github.com/25dce026-krishna/StudentHub

Learning Outcomes:

- Constructed a complete data-driven Student CRUD management system in PHP and MySQL.
- Implemented secure parameterized prepared statements across all database queries.
- Created multi-parameter search algorithms supporting name, email, and course filtering.
- Guarded against accidental data loss via client-side JavaScript confirmation dialogs.
- Designed a soft-delete architecture preserving historical and relational integrity.
- Built an automated CSV export utility using PHP stream handlers.
- Managed and validated operations against a dataset of 21 seeded records.